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Research on Bayesian Optimization Algorithm Selection Strategy

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***Abstract*-Probability model accuracy is the base of Bayesian Optimization Algorithm and data sample is the base of construction accuracy model. So sample strategy is critical for the algorithm. In test, tournament selection,truncation selection and proportional selection are adapted to deal with typical dependency-free function, bivariate dependencies function and multivariate dependencies function. The result shows that tournament selection is the best selection strategy for Bayesian Optimization Algorithm, truncation selection and proportional selection are unsuitable for the algorithm.**

Index Terms --Bayesian Optimization Algorithm, tournament selection,truncation selection,proportional selection

I. INTRODUCTION

Estimation distribution algorithm was proposed in last decade and had been greatly developed in recent years. Statistical Learning is introduced into genetic algorithm to form estimation distribution algorithm. In EDA, probability model of individuals in the solution space is set up by statistical learning. Mutation operator in genetic algorithm is replaced by probability model, new population is sampled from probability model and new probability model based on new population is set up. The procedure is iterated to realize population evolution. Bayesian Optimization Algorithm is one kind of estimation distribution algorithm, it is proposed by American Pelikan in 2002. Bayesian network is used as probabilistic model to capture the (in)dependencies between the decision variables of the optimization problem in the algorithm. This probability model can reveal the nature of the problem. Since BOA is proposed, it is widely researched by many scholars all over the world.

In the past years, researchers focus on how to set up accurate probability model to increase efficiency of BOA. Mark and Pelikan combined prior problem information into graph[1]. Wu Yan adapted local structure learning to promote reliability, not increasing time complexity [2].But in fact, the probability model is set up based on data sample. Only when data sample shows the problem information accurately, the probability model can be accurate. So selection strategy is critical for the algorithm. Tournament selection, truncation selection, proportional selection are widely used in evolution algorithm, these methods are adapted in BOA respectively in experiment. From the result, analysis on which is the best selection strategy for BOA is discussed.

II. BAYESIAN OPTIMIZATION ALGORITHM

The first population of candidate solutions in BOA is typically generated at random according to the uniform distribution over all possible strings. Each iteration of BOA starts by selecting a new population of promising candidate solutions from the current population. Any selection methods used in population-based evolution algorithms can be used; for example, binary tournament selection, truncation selection and proportional selection all can be used. Then, Bayesian network is set up based on the selected solutions. New solutions are generated by sampling the probability distribution encoded by the learned Bayesian network. Finally, the new solutions are incorporated into the original population; even more, this can be done by replacing the entire old population with the new solutions. The procedure is terminated when some predefined termination criteria are reached; for example, when a solution of sufficient quality has been reached or when the population has lost diversity and it is unlikely that BOA will reach a better solution than the solution that has been found already[3]. The procedure of BOA is outlined as follow:

Bayesian optimization Algorithm(BOA)
1) set t:=0, randomly genetic initial population P(0)
2) select a set of promising strings S(t) from P(t)
3)construct the network B using a chosen metric and constrains
4)generate a set of new strings O(t) according to the joint distribution encoded by B
5)create a new population P(t+1) by replacing some strings from P(t) with O(t),set t:=t+1
6)if the termination criteria are not met, go to (2)

In BOA, the promising solutions are selected for the current population and probability model revealing the distribution of the solution is set up according these selected solutions, then sample randomly from the built model to get new population. In the procedure, it is obviously how to select the promising solution directly decide the probabilistic model and then the new population and the algorithm evolution process.

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III. SELECTION OPERATOR

Tournament selection, truncation selection and proportional selection are used widely in evolution algorithm[4].

1) Tournament selection

In tournament selection, T_{our} individuals are randomly selected from population, then the best one is selected as the parent individual. The procedure is iterated to get all selected individuals. The parameter of tournament selection is T_{our}, it is belong to [2, Nind].

2) Truncation selection

In truncation selection, the individuals are sorted as fitness value. Only the best one is selected as parent individual. The parameter of truncation selection is T_{runc}, which is defined as the percentage of selected parent individual and between 10% and 50%. The individuals less than T_{runc} can not produce offspring.

3) Proportional selection

In proportional selection, the individual selection probability is proportional to its fitness value as following:

$$P_i = \frac{f_i^k}{\sum_{i=1}^{M} f_i^k} \tag{1}$$

f_i is the fitness value for individual i. And k is a random positive integer. Lesser k is, lesser the selection pressure is. In this situation the worse individual has some survival chance, the searching space is large too. Bigger k is, larger the selection pressure is. In this situation, the worse individual has no survival chance, the searching space is small too. When k is large, selection is made within several large fitness value individuals.

IV. EXPERIMENT

A. Test Function

Three typical functions are tested in this experiment, which are described as follow.

Function 1:onemax

Onemax is defined as the sum of every bit of input string, the mathematical expression is as follow.

$$f_{one\max}(X) = \sum_{i=0}^{n-1} x_i \tag{2}$$

This function gets its maximal when every bit of the input string is one. It is a monotonous single peak function, there is no relation between variables. The function is easy to be optimized.

Function 2: 3-order deceptive function

$$f_{deceptive}(x) = \begin{cases} 1.0 & if \quad u = 2 \\ 0.9 & if \quad u = 0 \\ 0.0 & if \quad u = 1 \end{cases} \tag{3}$$

There, $u(x) = \sum_{i=0}^{l-1} x_i$, and x is a binary string, x_i is for the ith bit. This is a bivariate function. In experiment, the test function is composed of n 3-order deceptive functions, which can be expressed as follow.

$$F(X) = \sum_{i=0}^{m-1} f_{deceptive}(u_i) \tag{4}$$

Function 3: 5-order trap function

$$f_{trap}^5(X) = \begin{cases} 4-u & if \quad u < 5 \\ 5 & otherwise \end{cases} \tag{5}$$

There, $u(x) = \sum_{i=0}^{l-1} x_i$, x stands for binary string, x_i stands for the ith bit. This is a multivariate dependencies function. In experiment, the test function is composed of n 5-order trap functions, which can be expressed as follow.

$$F(X) = \sum_{i=0}^{m-1} f_{trap}^5(u_i) \tag{6}$$

B. Method of Experiment

Tournament selection, truncation selection and proportional selection are adapted in BOA respectively on testing function 1, function 2 and function 3 in experiment. Bayesian network is constructed all from empty network in every generation in BOA. In tournament selection, T_{our}=4. In truncation selection, T_{runc}=50%. In proportional selection, two methods are adapted. One is fixed proportion, k=1. The other is varying proportion, k is self-adaptive to selection pressure.

Evaluation is a critical index to judge algorithm performance. So population scale, evolution generation and fitness evaluation are selected as criteria. Several independency experiments are taken, and the best result are selected.

C. Experiment Data

1) On experiment adapted tournament selection and truncation selection respectively in BOA for function 1, problem size increases from 10 to 30, experiment data are showed in table I.

TABLE I
EXPERIMENT DATA ON FUNCTION 1

problem size	population scale		convergent generation		fitness evaluation	
	tourn	trunc	tourn	trunc	tourn	trunc
10	200	300	2	8	300	1500
15	200	400	3	5	500	1400
20	200	40000	10	49	1200	1020000
25	400	100000	6	28	1600	1500000
30	400	120000	8		2000	

When problem size increases to 30, using truncation selection, population scale even enlarges to 120000, the optimal solution can not be found even after 100 generations. Using tournament selection, more large scale problem can be solved. For example, when problem size increases to 90, population scale is 1000, the best optimal solution can be found in 14th generation. The evaluation is 8000.

2）On experiment adapted tournament selection and truncation selection respectively in BOA for function 2, problem size increase from 12 to 30. Experiment data is showed as table II.

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TABLE II
EXPERIMENT DATA ON FUNCTION 2

problem size	population scale		convergent generation		fitness evaluation	
	tourn	trunc	tourn	trunc	tourn	trunc
12	400	400	3	7	1000	1800
15	400	600	2	22	800	7200
18	400	2100	4	13	1200	15750
21	600	12000	8	24	3000	156000
24	600	100000	8		3000	
30	600		9		3300	

When problem size increases to 24, using truncation selection, population scale increases to 100000, the optimal solution can not be found even after 100 generation. Adapted tournament selection, more large scale problem can be solved. For example, when problem size increases to 90, population scale is 3000, the optimal solution can be found in 19th generation. The evaluation is 31500.

3）On experiment adapted tournament selection and truncation selection respectively in BOA for function 3, the problem size increase from 10 to 25. The experiment data is showed as table III.

TABLE III
EXPERIMENT DATA ON FUNCTION 3

problem size	population scale		convergent generation		fitness evaluation	
	tourn	trunc	tourn	trunc	tourn	trunc
10	200	500	2	5	400	1750
15	200	3000	3	4	500	9000
20	400	40000	9	49	2200	1020000
25	600	60000	12	47	4200	1470000
30	800	180000	13		6000	

When problem size increases to 30, using truncation selection, population scale increases to 180000, the optimal solution can not be found even after 100 generation. Adapted tournament selection, more large scale problem can be solved. For example, when the problem size increases to 90, population scale is 8000, the optimal solution can be found in 14th generation. The evaluation is 64000.

4) Adapted proportional selection in BOA on testing function 1, when problem size is 10 and the population scale is 200, the optimal solution can be found in the 12th generation and the evaluation is 1400. when problem size is 15 and the population scale is 1000, the optimal solution can be found in the 4th generation and the evaluation is 3000. But when problem size is up to 20, population scale is 100000, the optimal solution can not be found even after 100 iterations. Adapted proportional selection in BOA on testing function 1 and function 3, the optimal solution can not be found when k equals 1 or k is self-adaptive in the evolution process.

5) On experiment adapted truncation selection respectively in BOA for function 2 and function 3, the evolution process is reversal. Experiment result is showed as figure 2 and figure 3.

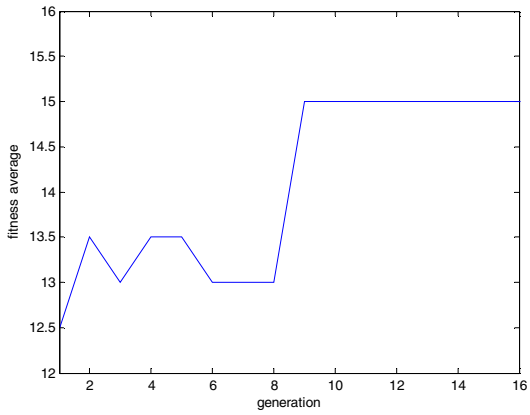


Fig.1 Evolution process of function 2

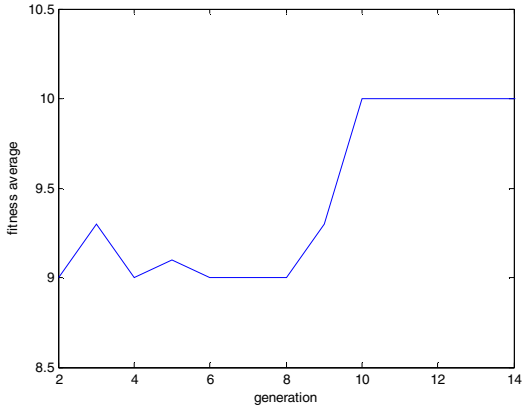


Fig.2 Evolution process of function 3

D. Experiment Analysis

From the experiment result of three selection strategy adapted in BOA on testing three typical functions, analysis is as follow.

1) Adapted tournament selection in BOA, BOA performs steady, the needed population scale is the smallest, the evaluation is the least too, and algorithm performance is the best. In tournament selection, the offspring individuals are the best ones from the data sample randomly selected from population. In this way, the variables relationship can be retained. So the probability model can reflect the relationship of variables. More accurate the model is, much better BOA performs.

2) Adapted truncation selection in BOA, BOA does not perform steady, the needed population scale is larger, the evaluation is larger too, and algorithm performance is worse. In truncation selection, individual is ranked according to its fitness value, and the individual having high fitness value is kept, those individuals having lower fitness value are discarded. Then the variables relationship containing in those discarded individuals is lost, so the probability model set up based on these incomplete data can not reflect variables relationship correctly, so the searching process is full of reversals and zigzags. Even more, the optimal solution can not be found.

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3) Adapted proportional selection in BOA, even population scale is very large, it is difficult for BOA to find the optimal solution. The proportion of individual fitness value decides whether the individual is kept or discarded. It is obvious that higher the selection probability the individual has, the more times the individual is kept. So selection is done within several individuals having high fitness value. The selected individuals in this way obviously can not reflect the variables relationship, and probability model is not accurate too. BOA adapted this selection strategy can not run effectively obviously.

CONCLUSION

It is critical for BOA to setting up accurate probability model. Keeping data sample validity is the essential prerequisite for setting up accurate probability model. How to select the individuals which contain the variables relationship from population will directly affect algorithm effectiveness. Experiment on three typical test functions adapting different selection operator has been done. The experiment result and analysis show that tournament selection is the best selection strategy for BOA, proportional selection and truncation selection are not suitable for BOA. Theoretic study and application on engineering problems are the further research.

REFERENCES

[1] Mark Hauschild, Martin Pelikan, Kumara Sastry, and David E. Goldberg[R], Using previous models to bias structural learning in the hierarchical BOA, MEDAL Report No. 2008003, January 2008

[2] WU Yan, WANG Yu ping, LIU Xiaoxiong [J]. Improved bayesian optimization algorithm incorporated local structure learning. Systems Engineering and Electronics. Vol.30 No.12,p2493~2496, 2008

[3] Pelikan, M[D]. Bayesian Optimization Algorithm: From Single Level to Hierarchy.Ph.D. Dissertation from the Dept. of Computer Science at the University of Illinois at Urbana-Champaign 2002

[4] Wang Xiaoping, Cao Liming [M]. Genetic algorithm — — Therory, Applicantion and Software Implement. Xi'an. Xi'an Jiaotong University Press. 2002. P28~33, 2002

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